



CSE110: Object Oriented Programming [Spring 2024]

Project Report Pharmacy Management System

Submitted To

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1. Introduction:

The Pharmacy Management System is a software application designed to streamline and enhance the management of a pharmacy. It provides functionalities for both administrators (admins) and customers, allowing efficient management of medicines, orders, and payments. Admins can perform tasks such as adding, removing, and updating medicines, while customers can place orders and make payments.

2. Objectives:

The primary objectives of the Pharmacy Management System are:

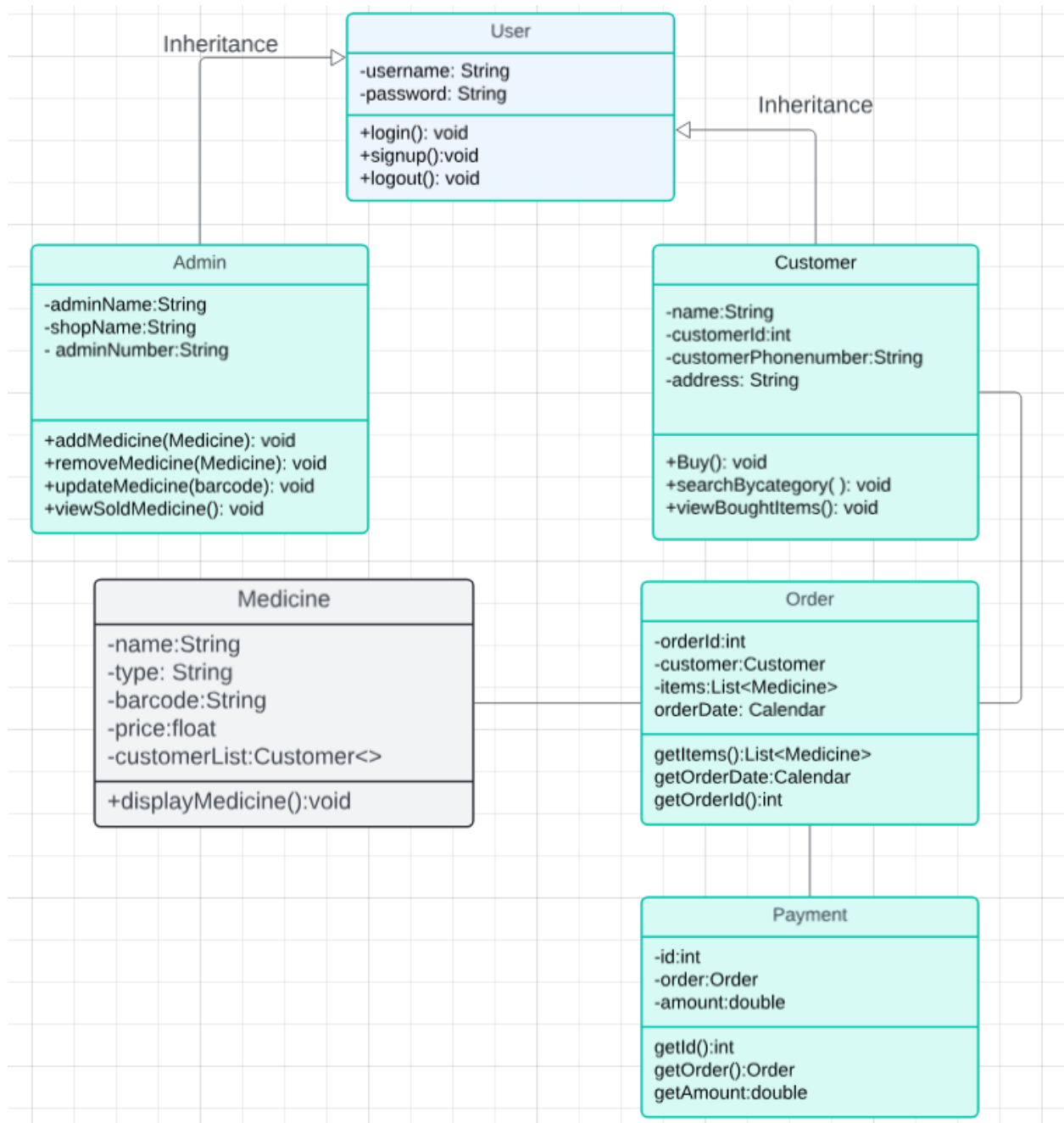
- To simplify pharmacy operations through a user-friendly interface.
- To provide secure user authentication for both admins and customers.
- To enable efficient management of the pharmacy's medicine inventory.
- To facilitate seamless order placement and payment processing for customers.

3. System Design:

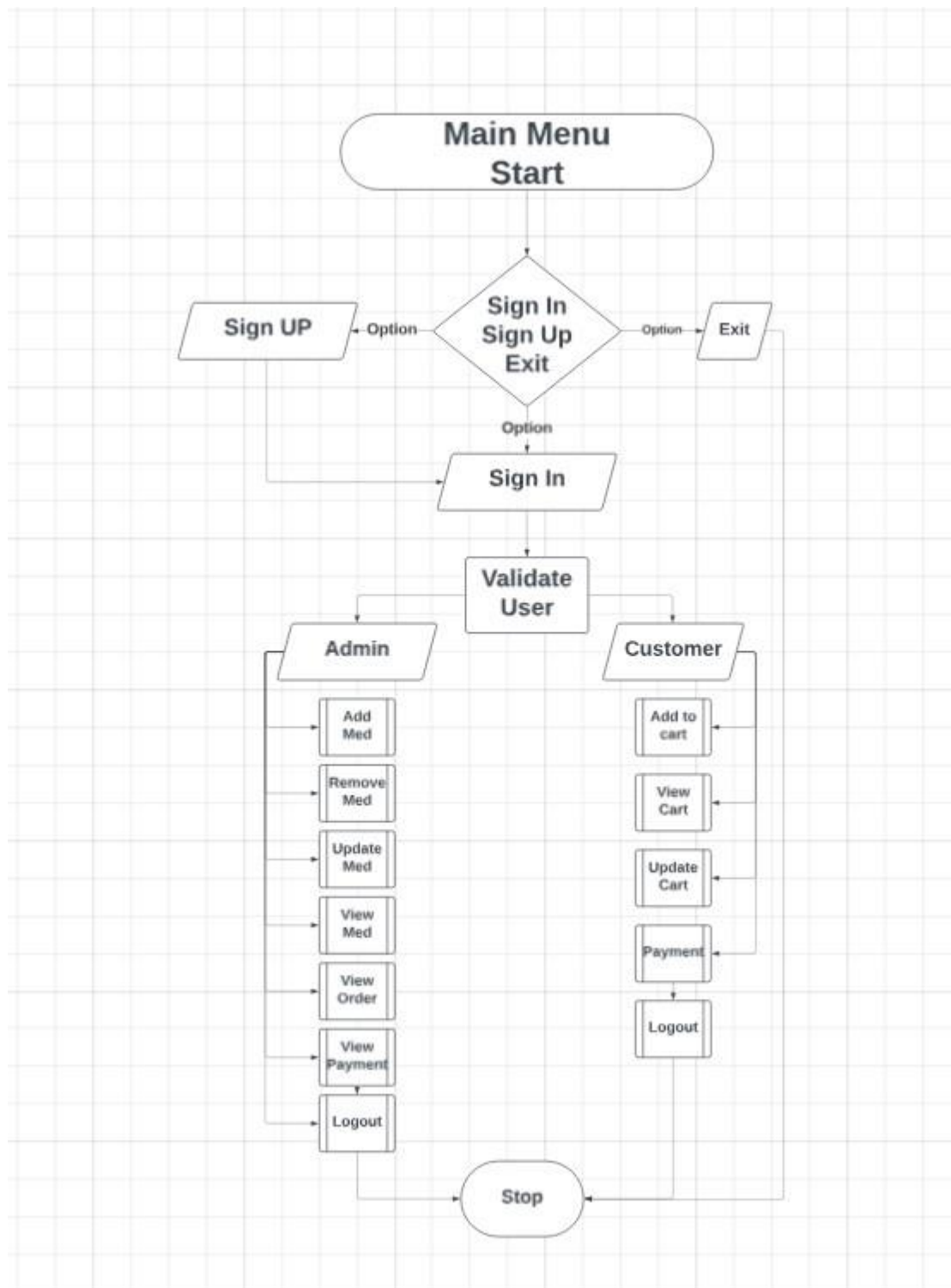
The system is designed using Java programming language and object-oriented principles. It consists of several classes representing different entities and functionalities within the pharmacy management domain:

- **User:** Base class for both Admin and Customer.
- **Admin:** Manages medicine inventory (add, remove, update, view).
- **Customer:** Places orders and makes payments.
- **Medicine:** Represents medicines available in the pharmacy.
- **Order:** Represents a customer's order.
- **Payment:** Handles the payment process for an order.

4. Uml Diagram:



5. Flowchart:



6. Functionalities:

User Features:

- **Sign up:**
- **Sign in:**

```
Welcome to the Pharmacy Management System!
```

```
Main Menu:
```

```
1. Sign Up
```

```
2. Sign In
```

```
3. Exit
```

```
Enter your choice:
```

Admin Features:

```
Admin Menu:
```

```
1. Add Medicine
```

```
2. Remove Medicine
```

```
3. Update Medicine
```

```
4. View Medicines
```

```
5. View Orders
```

```
6. View Payments
```

```
7. Logout
```

```
Enter your choice: █
```

- **Add Medicine:** Allows adding new medicines to the inventory.
- **Remove Medicine:** Enables removal of medicines from the inventory.
- **Update Medicine:** Allows updating details of existing medicines.
- **View Medicines:** Displays the current inventory of medicines.
- **View Orders:** Displays all the order which has taken by a customer.
- **View Payment:** Allows to check the pas history of transaction.

Customer Features:

```
Customer Menu:
```

```
1. Add to Cart
```

```
2. View Cart
```

```
3. Update Cart
```

```
4. Pay
```

```
5. Logout
```

```
Enter your choice: █
```

- **Add to cart:** Able to add medicine in the cart.
- **View cart:** Display the added item.

- **Update cart:** Able to make change in the cart.
- **Payment:** Generate the bill.

```
-----  
Total Price: Tk.14.98  
Enter the payment amount: Tk.14.98  
Payment successful. Change: Tk.0.0  
Your cart is now empty.
```

7. Implementation

The system is implemented in Java, using separate classes for each entity and functionality. The Main class handles user interactions, including sign-up, sign-in and menu navigation for admins and customers. The system utilizes object-oriented concepts for modularity and code reusability.

Class Structure:

User: Base class for Admin and Customer.

Admin: Inherits from User, manages medicine inventory.

Customer: Inherits from User, places orders and makes payments.

Medicine: Represents medicines with attributes such as ID, name, price, and quantity.

Order: Represents an order with customer details, items, and date.

Payment: Represents payment with order details, amount and billing.

8. Conclusion:

The Pharmacy Management System provides a robust solution for managing pharmacy operations efficiently. It automates administrative tasks and simplifies the order and payment processes for customers. The system is designed with scalability and maintainability in mind, allowing for future enhancements such as a graphical user interface and database integration.

This report summarizes the project output and functionalities of the Pharmacy Management System, emphasizing its usability and the implementation details in Java.